

Real-Time Interpretable Monocular Motion Forecasting and Reactive Navigation for Autonomous Surface Vehicles

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Abstract—Autonomous surface vehicles (ASVs) usually infer their own motion from GNSS and inertial sensing, both of which can be denied, degraded or simply unavailable on small low-cost platforms. We present a complete monocular pipeline that estimates vessel ego-motion, forecasts the short-horizon trajectory, builds a local risk representation and issues an interpretable navigation command, using nothing but a forward-facing RGB camera and no GPS, map, depth sensor or absolute scale. The system runs end-to-end at 70FPS on commodity hardware. Beyond describing the architecture, our main contribution is an evaluation protocol that brackets the forecast between an explicit skill floor (zero-motion), a competitive baseline (constant velocity) and a hindsight oracle, rather than reporting self-consistency statistics alone. On 177s of real ASV footage the forecast attains a final displacement error of 13.1% and 13.7% of the distance travelled at 1s and 2s horizons, improving on the zero-motion floor by 86%. The same protocol exposes results that a conventional evaluation would have concealed: the constant-turn model yields no benefit over a straight constant-velocity extrapolation, the yaw signal driving path curvature is measured on the wrong image channel ($R^2 = 2\%$ against an independent reference, versus 84% for a corrected channel), and near-field bow wake saturates the risk map so that the safety filter never activates. We report these negative findings with the measurements that establish them, and argue that bracketed baselines should be standard practice for vision-only maritime motion forecasting.

Index Terms—Autonomous surface vehicles, optical flow, trajectory prediction, monocular vision, interpretability, reinforcement learning, maritime robotics.

I. INTRODUCTION

Reliable short-horizon motion estimation underpins every reactive behaviour an autonomous surface vehicle performs, from station keeping to collision avoidance. The dominant solution couples GNSS with an inertial measurement unit and, increasingly, AIS-derived context [1], [2]. That solution is effective but fragile: satellite positioning can be jammed or spoofed, inertial drift compounds during outages, and the sensor budget is frequently unavailable on the small, inexpensive platforms where autonomy would be most transformative.

This motivates a complementary question. *How much navigational competence can be recovered from a single camera alone?* A forward-facing RGB sensor is cheap, passive, and already present on most vessels for situational awareness. If ego-motion, a short-horizon forecast and a usable risk representation

can be derived from the image stream in real time, the resulting estimate degrades independently of the satellite constellation and can serve either as a fallback or as a cross-check.

A second requirement follows immediately. Maritime autonomy is a safety-critical, regulated domain, so a system that cannot justify its commands is difficult to certify and harder to trust. We therefore treat *interpretability* as a design constraint rather than a post-hoc analysis step [3], [4]: every stage of the pipeline emits a quantity a human can inspect, and every command is accompanied by the evidence that produced it.

Our previous work established that dense optical flow sampled near the image centre provides a stable proxy for instantaneous ego-motion and can be linearly extrapolated over a short horizon. That study, in common with much of the vision-only literature, reported metrics — step length, heading alignment, coverage — that characterise the *internal consistency* of the predictor. Such metrics cannot answer the question a practitioner actually asks: is the prediction better than assuming nothing changes, and is it better than the simplest alternative? The present paper closes that gap and extends the system from perception into decision making. Our contributions are as follows.

- 1) A complete, real-time (70 FPS), GPS-free monocular pipeline spanning ego-motion estimation, short-horizon forecasting, residual-flow risk mapping, policy-based action selection, a safety filter, and per-frame natural-language explanation (Sec. III).
- 2) An evaluation protocol that brackets the forecast between a zero-motion skill floor, a constant-velocity competitor and a hindsight oracle, applied to real ASV footage (Sec. V-A).
- 3) A controlled comparison of a classical potential-field controller against PPO and Double DQN policies over an identical observation and reward, with an observation ablation (Sec. V-C).
- 4) A diagnosis of two concrete failure modes that the bracketed protocol exposes — a mis-assigned yaw channel and risk-map saturation by near-field wake — each quantified against an independent reference (Sec. V-D).

II. RELATED WORK

Maritime motion estimation. Classical ASV state estimation fuses GNSS and inertial measurements, with recent work

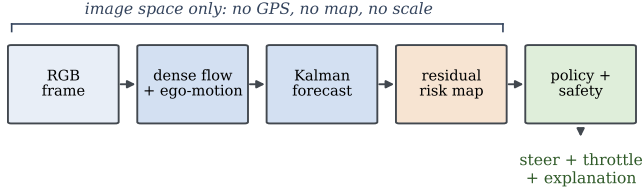


Fig. 1. System overview. Everything left of the policy block operates purely in image space: no GPS, no map, no depth and no absolute scale. Each stage emits an inspectable quantity, and the predicted heading is the common frame of reference shared by the risk map and the command.

improving robustness through invariant filtering [1]. Model-based approaches such as Nomoto or Fossen formulations require hydrodynamic parameters that are difficult to identify and vary with sea state [5]. Vision offers a parameter-free alternative, and electro-optical processing for maritime object detection and tracking is by now a mature field [6], [7].

Optical flow. Dense formulations descend from the variational treatment of Horn and Schunck [8]; Farnebäck’s polynomial expansion [9] remains a robust classical choice, while dense inverse search (DIS) [10] attains comparable accuracy at a fraction of the cost, which matters for onboard deployment. Learned estimators such as RAFT [11] are more accurate on standard benchmarks but assume compute we deliberately exclude. In maritime scenes dense flow is generally preferred over sparse tracking because stable visual features are scarce and the background is highly dynamic [6]. Sparse tracking nevertheless remains valuable for global motion: Shi–Tomasi corners [12] tracked by Lucas–Kanade [13] and fitted with RANSAC [14] give a robust affine estimate of camera motion, which we use to complement the dense field.

Trajectory prediction. Data-driven vessel forecasting from AIS has advanced rapidly [2], and pedestrian-motion architectures such as Social LSTM [16] popularised sequence models for trajectory tasks. These methods target horizons of minutes to hours and consume historical position tracks; they do not address the sub-second visual loop required for immediate kinetic stability, and they presuppose the positioning that we assume absent.

Reactive navigation. Potential fields [17] and the dynamic window approach [18] remain the reference reactive planners because their behaviour is transparent and bounded. Learned alternatives based on PPO [19] or value-based methods in the DQN family [20]–[22] promise better asymptotic performance at the cost of interpretability. Collision avoidance for vessels carries the additional burden of COLREGs compliance [23]. We compare the classical and learned options under identical inputs and rewards, and report the comparison faithfully.

III. METHOD

Fig. 1 summarises the pipeline. Frames are resized to 512×512 and converted to greyscale; all quantities are expressed in pixels per frame.

A. Ego-motion Estimation

Dense flow $\mathbf{F}_t$ is computed between consecutive greyscale frames using DIS [10] with spatial propagation and variational refinement, falling back to Farnebäck [9] where unavailable.

Two independent estimates of camera motion are then formed. The first tracks Shi–Tomasi corners [12] with Lucas–Kanade [13] and fits a partial affine transform $\mathbf{M}$ by RANSAC [14], from which the displacement at the image centre follows. The second reduces the dense field inside a central region of interest to a single vector using a median/MAD outlier rejection with Gaussian spatial weighting: with $\mu = \text{med}(\mathbf{F})$ and $\sigma = \text{med}(\|\mathbf{F} - \mu\|)$, samples beyond $k\sigma$ ($k = 2.5$) are discarded before the weighted mean is taken. Writing ρ_a and ρ_d for the inlier ratios of the two estimates, the fused ego-velocity is

$$\mathbf{v}_t = \frac{\rho_a \mathbf{v}_t^{\text{aff}} + \rho_d \mathbf{v}_t^{\text{den}}}{\rho_a + \rho_d}, \quad (1)$$

so an estimate that fails to reach consensus contributes proportionally less.

B. Filtering and Forecast

The fused velocity drives a constant-acceleration Kalman filter [15] with state $\mathbf{x} = [v_x, v_y, a_x, a_y]^T$. Measurement noise is adaptive, $\mathbf{R} = r\rho^{-1}\mathbf{I}$, so the filter discounts the flow exactly when consensus deteriorates.

Because the camera is bow-mounted the vessel always advances into the scene. The forecast integrates a constant-turn model forward over $N = \lceil Hf \rceil$ frames for horizon H at frame rate f , with heading offset $\theta_0 = \text{clip}(\kappa v_x)$ and turn rate ω :

$$\theta_k = \theta_{k-1} + \omega_{k-1}, \quad \omega_k = \lambda \omega_{k-1}, \quad \lambda = 0.97, \quad (2)$$

$$p_k^x = p_{k-1}^x + \|\mathbf{v}\| \sin \theta_k, \quad p_k^y = p_{k-1}^y - \|\mathbf{v}\| \cos \theta_k, \quad (3)$$

where λ damps the turn over the horizon rather than assuming it persists. A confidence score $c = \frac{1}{2}\rho + \frac{1}{2}s$ combines the inlier ratio with heading stability s and sets the width of the displayed uncertainty cone.

C. Residual Flow and Local Risk

Subtracting the fitted global motion from the dense field leaves the *residual flow*, $\mathbf{F}^{\text{res}} = \mathbf{F} - \mathbf{M} \cdot \mathbf{p}$, which isolates motion the vessel is not itself causing. Its magnitude is pooled into a 5×5 grid, normalised and clipped to $[0, 1]$; cells at or above 0.6 are flagged high risk. A directional projection weights cells by their alignment with a query heading, yielding risk ahead, to port and to starboard.

D. Observation, Actions and Policies

All controllers consume the same fixed 31-dimensional observation interface as presented in Table I. The first two entries can be interpreted as the forecast intent encoding, where the predicted turn/tilt, indicates how far the forecast heading deviates from straight-ahead and the normalized image-space speed of the vessel. Entries 2–4 transform the residual-flow cost map into directional risks which are evaluated in the forecast heading frame. Thus the policy sees risk in relation to the

TABLE I
OBSERVATION VECTOR: ALL ENTRIES ARE CLIPPED TO FIXED RANGES BEFORE POLICY INFERENCE.

Index	Quantity	Interpretation
0	predicted tilt $[-1, 1]$	forecast heading deviation: left/port < 0 , right/starboard > 0
1	normalized speed $[0, 1]$	image-space ego speed
2–4	risk ahead/left/right $[0, 1]$	directional projections of residual-flow grid
5	temporal stability $[0, 1]$	consistency of recent forecast headings
6–30	risk grid $[0, 1]$ ²⁵	flattened 5×5 residual-flow cost-map

path it is expected to follow instead of associating it to fixed image axes. Next, temporal stability provides the policy with a confidence similar signal for the recent heading consistency, determining motion estimation in the process. The remaining 25 entries preserve the full spatial residual-risk layout, therefore allowing the learned policies to condition on asymmetry, near-field saturation and residual motion influenced by obstacles, beyond the three directional summaries.

Thus, every entry is range-bounded so the learned policies see a fixed scale. Actions factorise into steering (five detents from hard port to hard starboard) and throttle (decelerate, hold, accelerate). The shared reward

$$r = -1.5 \eta_{\text{sel}} + 0.3 \phi_{\text{thr}} + 0.15 \phi_{\text{str}} - \psi_{\text{smooth}} \quad (4)$$

penalises the risk η_{sel} along the chosen bearing, credits agreement with the desired throttle and steering (ϕ terms, graded by ordinal distance), and discourages abrupt action changes.

Three controllers are compared. **Potential fields** [17] form a steering signal from the predicted turn and the port–starboard risk difference and act past a fixed threshold. **PPO** [19] uses a shared two-layer body with separate steering, throttle and value heads. **Double DQN** [21] learns Q over the 15 joint actions with a target network and prioritised replay [22]; we additionally weight sampling by the risk observed at each stored frame, since high-risk frames are rare in maritime footage and would otherwise be swamped.

E. Safety Filter and Explanation

An independent filter caps the command whenever risk ahead exceeds 0.6: it forces deceleration, steers toward the clearer side, and marks the decision as overridden. Crucially both the raw policy action and the filtered action are retained, so intervention is always visible. Each frame additionally emits a natural-language justification and a JSONL record containing the forecast, cost map, observation, both actions, reward and timing, enabling post-hoc audit of any moment in a run.

IV. EXPERIMENTAL SETUP

Experiments use 176.7 s of real ASV footage recorded from a bow-mounted camera on an inland waterway (640×400 at 21.9 FPS, 3867 frames), provided by a project partner¹. The clip contains open-water transit, a navigation buoy and a vessel ahead. Two additional sample videos are used for policy training. Timing is measured on a commodity x86 CPU with no GPU acceleration. Policies are trained for 40k environment steps; all reported policy results are from replay of held-out footage.

¹Dataset provenance anonymised for review.

TABLE II
FORECAST ACCURACY ON REAL FOOTAGE ($n \approx 3.8\text{k}$ WINDOWS PER HORIZON).
FDE/ADE ARE FINAL AND AVERAGE DISPLACEMENT ERROR. RELATIVE FDE IS EXPRESSED AS A FRACTION OF THE DISTANCE ACTUALLY TRAVELLED.

H (s)	Proposed			Baseline FDE (px)			Skill
	FDE	ADE	rel.	zero	c.vel.	oracle	
0.5	1.56	0.82	12.6%	11.9	1.56	0.03	+87%
1.0	3.24	1.65	13.1%	23.7	3.22	0.07	+86%
1.5	4.93	2.48	13.2%	35.5	4.86	0.12	+86%
2.0	6.83	3.36	13.7%	47.2	6.70	0.17	+86%

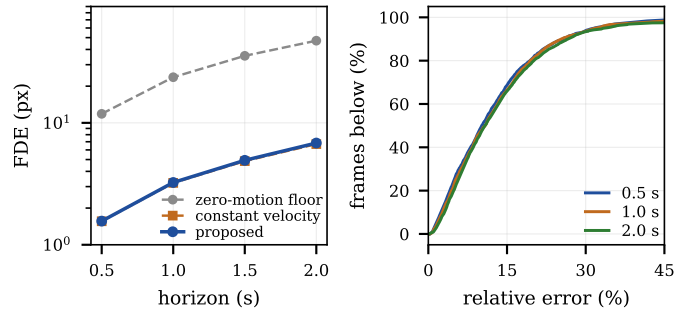


Fig. 2. Left: error growth against the bracketing baselines (log scale). The proposed forecast tracks the constant-velocity curve almost exactly and stays far below the zero-motion floor. Right: cumulative distribution of relative endpoint error; the median frame is accurate to roughly 11% at every horizon.

V. RESULTS

A. Forecast Accuracy

At every frame we roll the forecast forward over the horizon and compare it against the same kinematic model driven by the motion actually measured on the subsequent frames, so prediction and reference share units and differ only in that one is causal. Three baselines bracket the result: *zero-motion* (predict no displacement) is the skill floor, *constant-velocity* (current speed and heading, no turn) is the competitive alternative, and a *hindsight oracle* (mean speed and heading over the realised window) bounds what any constant-parameter model could achieve.

Table II and Fig. 2 report the outcome. The forecast is substantially better than assuming nothing moves — an 86% reduction in endpoint error that holds across horizons — and relative error grows only from 12.6% to 13.7% as the horizon quadruples, indicating that error accumulates roughly in proportion to distance rather than compounding. Median relative error is 10.6% at 1 s with a 90th percentile of 25.3%. Current speed predicts the mean speed of the next two seconds with $r = 0.85$.

TABLE III
PER-STAGE RUNTIME OVER 600 REAL FRAMES AT 512×512 , SINGLE CPU
THREAD, NO GPU.

Stage	Mean (ms)	p95 (ms)
Dense optical flow (DIS)	6.93	8.15
Global motion (LK + RANSAC)	3.77	4.80
Robust central velocity	0.75	0.95
Residual flow	3.03	3.79
End-to-end, potential fields	14.40	16.25
End-to-end, PPO	14.26	15.25
End-to-end, Double DQN	14.25	15.16

TABLE IV
CONTROLLERS ON REAL FOOTAGE UNDER AN IDENTICAL OBSERVATION AND
REWARD. MEAN RISK IS IDENTICAL ACROSS ROWS BECAUSE REPLAY IS
OPEN-LOOP; SEE TEXT.

Controller	Reward	Mean risk	Override	Switch rate
Potential fields	0.341	0.140	0.00	0.089
+ safety	0.341	0.140	0.00	0.089
PPO	0.292	0.140	0.00	0.571
+ safety	0.292	0.140	0.00	0.571
Double DQN	0.276	0.140	0.00	0.026
+ safety	0.276	0.140	0.00	0.026

The same table carries a result that self-consistency metrics cannot express: the proposed model does *not* improve on constant velocity. At 1 s it is 0.8% worse and at 2 s 2.0% worse. The Kalman filter and the constant-turn integration, which constitute most of the forecasting machinery, contribute nothing measurable over the assumption that the vessel simply continues as it is. We investigate why in Sec. V-D.

B. Real-Time Performance

Table III shows the pipeline closes the loop in 14.3 ms, or 70 FPS — over three times the 21.9 FPS source rate, leaving ample margin on an embedded target. Optical flow dominates at roughly half the budget. Policy inference is negligible: the three controllers differ by less than 0.2 ms, so the choice of decision model is not a real-time consideration.

C. Policy Comparison and Ablation

Both policies improve on the open-loop replay objective (Fig. 3): PPO rises from -0.24 to 0.185 mean reward per step over 40k steps, and Double DQN converges to the same level from -0.25 . Nevertheless Table IV shows the transparent potential-field controller achieving the highest reward on real footage, with PPO 14% and Double DQN 19% behind. This is expected because the potential-field rule structure closely matches the reward definition, meaning follow the predicted turn, prefer lower-risk side, decelerate under forward risk and avoid abrupt switches.

Three observations qualify this. First, the reward encodes preferences — a desired throttle and steering derived from risk and predicted turn — that the potential-field controller implements almost by construction, so it is close to optimal for this objective by design. Second, and more importantly, mean risk is *identical* across every row. Evaluation replays recorded footage open-loop, so a policy’s actions cannot alter

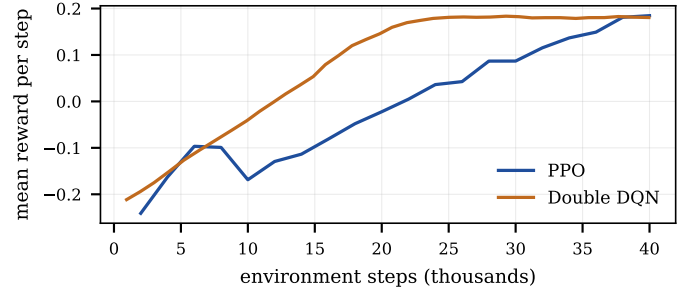


Fig. 3. Training curves. Both learners improve under the replay reward, demonstrating that the reward signal is learnable, although this does not indicate closed-loop navigation competence.

TABLE V
OBSERVATION ABLATION (PPO, TRAINED UNDER EACH VARIANT). TRAINING
RETURN SEPARATES THE VARIANTS; OPEN-LOOP EVALUATION DOES NOT.

Variant	Observation content	Train return	Eval reward
a1	intent + speed	0.080	0.337
a2	+ cost map	0.179	0.293
full	+ temporal stability	0.185	0.292

the observations it subsequently receives; the risk term is therefore constant and reward differences collapse onto the action-preference and smoothness terms alone. Third, the two learners adopt opposite degenerate strategies: PPO switches steering on 57% of frames while Double DQN switches on 2.6% and selects a forward command on only 1.1%.

Notably, the safety filter produces no overrides in this evaluation, thus the experiment does not validate whether the filter improves behaviour under genuine imminent risk. Currently, it exclusively verifies that the filter is applicable to be inserted without changing policy interface.

The ablation in Table V makes the protocol’s limitation explicit. Policies are trained and evaluated on open-loop replay, based on precomputed visual features. Thus, actions influence the immediate reward and previous smoothness term, but are not capable of affecting the subsequent visual state. Hence, results measure agreement with the action preference and the defined reward, and not a closed-loop navigation performance. The ablation indicates that residual-flow features assing in optimizing the replay reward as adding the cost map increases training from $0.080 \rightarrow 0.179$, and adding temporal stability provides a smaller gain to 0.185 . Nonetheless, the held-out replay ordering inverts because the a1 policy can collapse to a nearly constant action and avoid smoothness penalties. Thereby steering switch rate is shown as 0.000. A degenerate controller is competitive under open-loop replay. We regard this as the central methodological lesson of the study: *open-loop replay cannot validate a closed-loop controller*, and results obtained that way — including ours — should be read as action-preference statistics, not navigation performance.

D. Diagnostics on Real Data

Sec. V-A established that the constant-turn model adds nothing. The bracketed protocol lets us localise the cause. For a bow-mounted camera, vessel yaw translates the far field

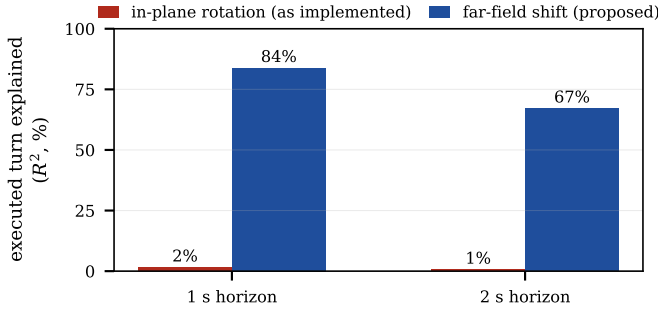


Fig. 4. Variance of the executed turn explained by the forecast. Curvature is currently driven by the affine in-plane rotation, which for a bow-mounted camera measures roll induced by wave motion. Reading the same quantity from the horizontal displacement of the far field — where yaw actually appears — recovers most of the signal.

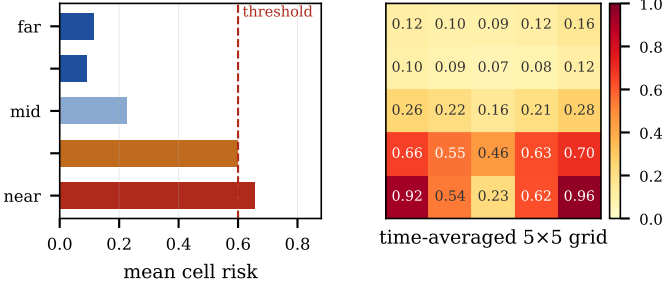


Fig. 5. Time-averaged residual-flow risk over the full clip. The two nearest grid rows sit at or above the high-risk threshold almost permanently, because they observe the vessel’s own bow and wake rather than the scene ahead.

horizontally across the image, whereas *roll* rotates the image in plane. The implementation derives its turn rate from the rotational component of the fitted affine transform — that is, from roll. Tracking features in the sky and shoreline band, which is effectively at infinity and therefore moves only under rotation, provides an independent reference. Against it, the quantity currently used explains $R^2 = 2\%$ of the turn actually executed. Extrapolating the far-field horizontal displacement instead, through the identical damped constant-turn model, explains $R^2 = 84\%$ at 1 s and 67% at 2 s (Fig. 4). The algorithm is sound; the input channel is not. The same analysis shows the damping constant systematically under-predicts, recovering 75% of the executed turn at 1 s and 58% at 2 s.

The risk map exhibits a second, independent failure. Over the full clip the maximum cell saturates at 1.0 in 87.1% of frames and at least one cell exceeds the high-risk threshold in 99.9% of frames, including long stretches of entirely empty water. Fig. 5 localises the cause: mean risk by row rises from 0.12 in the far field to 0.60 and 0.65 in the two nearest rows, which image the vessel’s own bow and wake. Because ego-motion subtraction removes a global affine transform, it cannot cancel the parallax of structure rigidly attached to the camera. The map is not blind to genuine obstacles: over the frames containing the navigation buoy and the vessel ahead, the mean far-field cell value rises from its clip-wide 0.14 to 0.42. The difficulty is that the permanently saturated near rows pin the maximum and render the high-risk cell count uninformative. The directional projection used by the

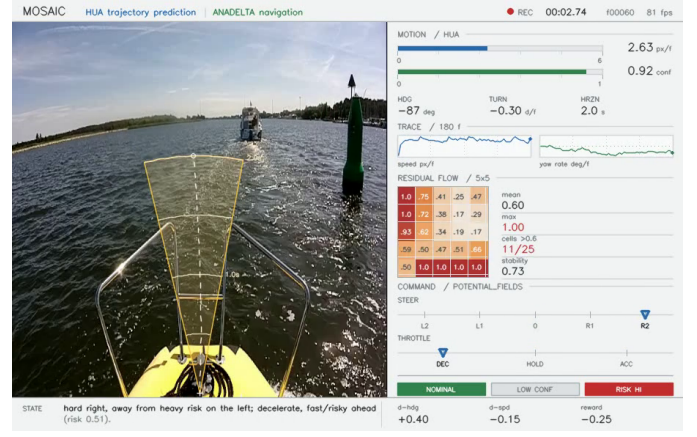


Fig. 6. Instrument view on real footage. The forecast envelope is drawn from the bow, its width encoding confidence and its arcs marking range at quarter-horizon intervals. The panel exposes motion estimates, the residual-flow grid, the commanded detents and the plain-language justification, so every command can be checked against the evidence that produced it.

safety filter weights distant cells and therefore ignores those near rows, with the consequence that the filter activated on 0.0% of frames across the entire recording. The safety layer is implemented and instrumented, but on this data it is never exercised.

E. Qualitative Behaviour

Fig. 6 shows the operator view during the approach to the navigation buoy. Mean forecast confidence over the clip is 0.897 (5th percentile 0.864), and the envelope visibly widens when flow consensus degrades. The figure also makes the saturation of Fig. 5 directly apparent: the lower grid rows read hot while the water ahead is clear.

VI. DISCUSSION AND LIMITATIONS

The perception front end delivers what it claims: a real-time, GPS-free, monocular forecast with 13–14% relative endpoint error on genuine ASV footage, fully instrumented and auditable. That result is solid and, to our knowledge, is the first vision-only maritime forecast reported against an explicit skill floor and a competitive baseline rather than against itself.

The remaining claims must be qualified. All quantities are in pixels per frame and carry no metric scale; without intrinsics or an independent reference the system cannot report m/s, and the pixel scale itself shifts by $1.36\times$ under the anisotropic resize from 640×400 to 512×512 . The curvature model is presently driven by the wrong signal, which both explains the absence of any gain over constant velocity and indicates a contained correction with measured expected benefit. The risk map requires either masking of the static bow region or a rescaling of the normalisation constant before the safety layer can function. Finally, the evaluation is open-loop, and Sec. V-C demonstrates that this protocol cannot discriminate between competent and degenerate policies; a closed-loop simulator or hardware-in-the-loop trial is necessary before any claim about learned navigation is warranted.

The recording itself is also a limitation: heading changes by no more than approximately 5° in any two-second window across the entire clip (the conversion from far-field pixel shift assumes a nominal field of view, absent intrinsics), so the data offers little opportunity for a turn model to demonstrate value. Footage containing deliberate manoeuvres, together with synchronised GNSS and inertial ground truth, is the single most valuable addition to this study.

VII. CONCLUSION

We presented a monocular, GPS-free perception and navigation pipeline for autonomous surface vehicles that runs end-to-end at 70FPS and exposes an inspectable quantity at every stage. Evaluated on real footage against explicit bracketing baselines, the short-horizon forecast reduces endpoint error by 86% relative to a zero-motion floor, achieving 13.1% and 13.7% relative error at one and two seconds.

The same protocol exposed three findings that a self-consistency evaluation would have hidden: the constant-turn machinery provides no benefit over constant velocity, because its yaw input is measured on a channel that carries roll ($R^2 = 2\%$ versus 84% for the corrected channel); near-field wake saturates the risk map so that the safety filter never activates; and open-loop replay cannot distinguish a trained policy from a degenerate one. We report these alongside the positive results deliberately. Bracketed baselines cost little and reveal a great deal, and we recommend them as standard practice for vision-only maritime motion forecasting.

Future work follows directly from the diagnostics: relocate the yaw estimate to the far-field channel, mask static vessel structure before risk pooling, and replace open-loop replay with closed-loop evaluation on manoeuvre-rich footage with inertial ground truth.

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REFERENCES

- [1] S. P. Singh, T. Lazouski, and M. Ghaffari, "GPS-drift: Marine surface robot localization using IMU-GPS fusion and invariant filtering," *arXiv preprint arXiv:2507.02198*, 2025.
- [2] N. Evmides, M. P. Michaelides, and H. Herodotou, "Vessel trajectory prediction with deep learning: Temporal modeling and operational implications," *Journal of Marine Science and Engineering*, vol. 13, no. 8, p. 1439, 2025.
- [3] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," *arXiv preprint arXiv:1702.08608*, 2017.
- [4] Y. Zhang, P. Tiño, A. Leonardis, and K. Tang, "A survey on neural network interpretability," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 5, no. 5, pp. 726–742, 2021.
- [5] S. Fujii, Y. Kawamura, and J. Tahara, "Modeling of μ -ASV for motion control using neural network," in *ISOPE International Ocean and Polar Engineering Conference*, 2021.
- [6] D. K. Prasad, D. Rajan, L. Rachmawati, E. Rajabally, and C. Quek, "Video processing from electro-optical sensors for object detection and tracking in a maritime environment: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 8, pp. 1993–2016, 2017.
- [7] R. Villarreal *et al.*, "An innovative methodology for segmenting vessel like structures using artificial intelligence and image processing," *Scientific Reports*, vol. 14, no. 1, p. 30332, 2024.
- [8] B. K. P. Horn and B. G. Schunck, "Determining optical flow," *Artificial Intelligence*, vol. 17, no. 1–3, pp. 185–203, 1981.
- [9] G. Farneback, "Two-frame motion estimation based on polynomial expansion," in *Scandinavian Conference on Image Analysis (SCIA)*, 2003, pp. 363–370.
- [10] T. Kroeger, R. Timofte, D. Dai, and L. Van Gool, "Fast optical flow using dense inverse search," in *European Conference on Computer Vision (ECCV)*, 2016, pp. 471–488.
- [11] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *European Conference on Computer Vision (ECCV)*, 2020, pp. 402–419.
- [12] J. Shi and C. Tomasi, "Good features to track," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1994, pp. 593–600.
- [13] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.
- [14] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.
- [15] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
- [16] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 961–971.
- [17] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, 1985, pp. 500–505.
- [18] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.
- [19] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [20] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [21] H. van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in *AAAI Conference on Artificial Intelligence*, 2016, pp. 2094–2100.
- [22] T. Schaul, J. Quan, I. Antonoglou, and D. Silver, "Prioritized experience replay," in *International Conference on Learning Representations (ICLR)*, 2016.
- [23] T. Statheros, G. Howells, and K. McDonald-Maier, "Autonomous ship collision avoidance navigation concepts, technologies and techniques," *Journal of Navigation*, vol. 61, no. 1, pp. 129–142, 2008.